Biopharmaceutical Science

Science

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Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

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Biopharmaceutical Science (A06313)

Short Title:	Biopharmaceutical Science		
Faculty	Science and Computing		
Department	Science		
Cluster	Pharmaceutical Science		
Author	AHEARNE		
Credits	5	**Level:**	Introductory

Description of Module / Aims

This module will provide an introduction to recombinant DNA technology and its relevance in the production of biopharmaceuticals. This will include the analysis of DNA structure and function, as well as the processes of cloning, screening and gene expression studies. The fundamental principles governing mechanisms of drug action and pharmacokinetics will also be provided.

Programmes

SCIE-0020	BSc (Hons) in Pharmaceutical Science (WD_SPHSC_B)	2 / 4 / M
SCIE-0020	BSc in Pharmaceutical Science (WD_SPHSC_D)	2 / 4 / M

Indicative Content

- Molecular genetics & biotechnology: DNA structure and function; DNA replication, transcription and translation; gene transfer and plasmids; restriction enzymes, vectors, cloning and screening; PCR; biopharmaceutical production by fermentation, harvesting and purification; monoclonal antibody production; gene therapy
- Microbiology: microbial nutrition and metabolism; microbial culturing and aseptic techniques; microbial growth; isolation of a pure culture; physical and chemical methods for controlling growth; action of antibiotics; antimicrobial drugs
- Immunology: the immune system; antibodies and interferon; immune tolerance and autoimmune disease
- The biopharmaceutical industry: growth and development of the industry; biopharmaceutical products; current/future trends; drug development and regulation
- Human physiology & pharmacology: introduce the basic concept of drug interaction and the major systems of the human body; absorption, distribution, metabolism/biotransformation and elimination of drugs; bioavailability of drugs

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Describe the basic technologies used to produce recombinant DNA.
2. Review the role of recombinant DNA technologies in the production of biopharmaceuticals.
3. Demonstrate how molecular biology has improved our understanding of many areas of biology and medicine.
4. Identify the potential of micro-organisms for antibiotic and pharmaceutical production.
5. Explain the rapid growth in the biopharmaceutical industry and discuss its control and future potential.
6. Discuss the fundamental principles of mechanisms of drug action and pharmacokinetics.
7. Use laboratory techniques to design simple strategies for the manipulation of DNA, examine common bacteria, assess antibiotic efficacy, utilise ELISA techniques and to identify micro-organisms as well as assess hygiene levels using rapid identification / detection methods.
8. Record results of laboratory work using a combination of formal written laboratory reports and worksheets.

Learning and Teaching Methods

- Lectures.
- Laboratory-based practicals.
- Self-directed study.

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	50%	1, 2, 3, 4, 5, 6
Continuous Assessment	50%	
Practical	50%	7,8

Assessment Criteria

<40%: Little or no demonstration of an understanding of the fundamentals of biopharmaceutical science. Lack of competence with associated laboratory skills & practices & an inability to carry out critical thinking.

40%-49%: Will show a limited understanding of the fundamentals of biopharmaceutical science. May have some difficulty with applying critical thinking and problem-solving. Will display some ability with associated laboratory skills & practices.

50%-59%: As well as the above, the student will show a good understanding of the fundamentals of biopharmaceutical science with demonstration of some critical thinking & problem-solving skills. Will display good competence with associated laboratory skills & practices.

60%-69%: In addition, will demonstrate more detailed knowledge of biopharmaceutical science, very good critical thinking skills & a high level of competence & efficiency in associated laboratory practicals.

70%-100%: All the above to an excellent level. In addition, will demonstrate a comprehensive knowledge & understanding of biopharmaceutical science & be able to discuss the area. Will display advanced competence & independence with associated laboratory skills & practices.

Note:

- In addition to the normal regulations, a minimum of 35% must be achieved in both the final exam and continuous assessment components of the module. In cases where continuous assessment has a practical component attendance of at least 75% in the practicals is mandatory.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	24	
Practical	24	
Independent Learning	87	

Essential Material

- Campbell, N.A., J.B. Reece, L.A. Urry, M.L. Cain, S.A. Wasserman and Minorsky, P.V., Jackson, R.B. _Biology_. 8th ed. San Francisco: Pearson, Benjamin / Cummings, 2008.
- Ho, R.J.Y. and M. Gibaldi. _Biotechnology and Biopharmaceuticals: Transforming Proteins_ _and Genes into Drugs_. 2nd ed. Hoboken, NJ: John Wiley & Sons, 2013.

Supplementary Material

- Tortora, G.J., B.R. Funke and C.L. Case. _Microbiology - An Introduction_. 10th ed. San Francisco: Pearson, 2010.
- Walsh, G. _Pharmaceutical Biotechnology: Concepts and Applications_. Chichester, England; Hoboken, NJ: John Wiley & Sons, 2007.

Requested Resources

- Lecture Room: Loose Seated
- Room Type: Biology Lab